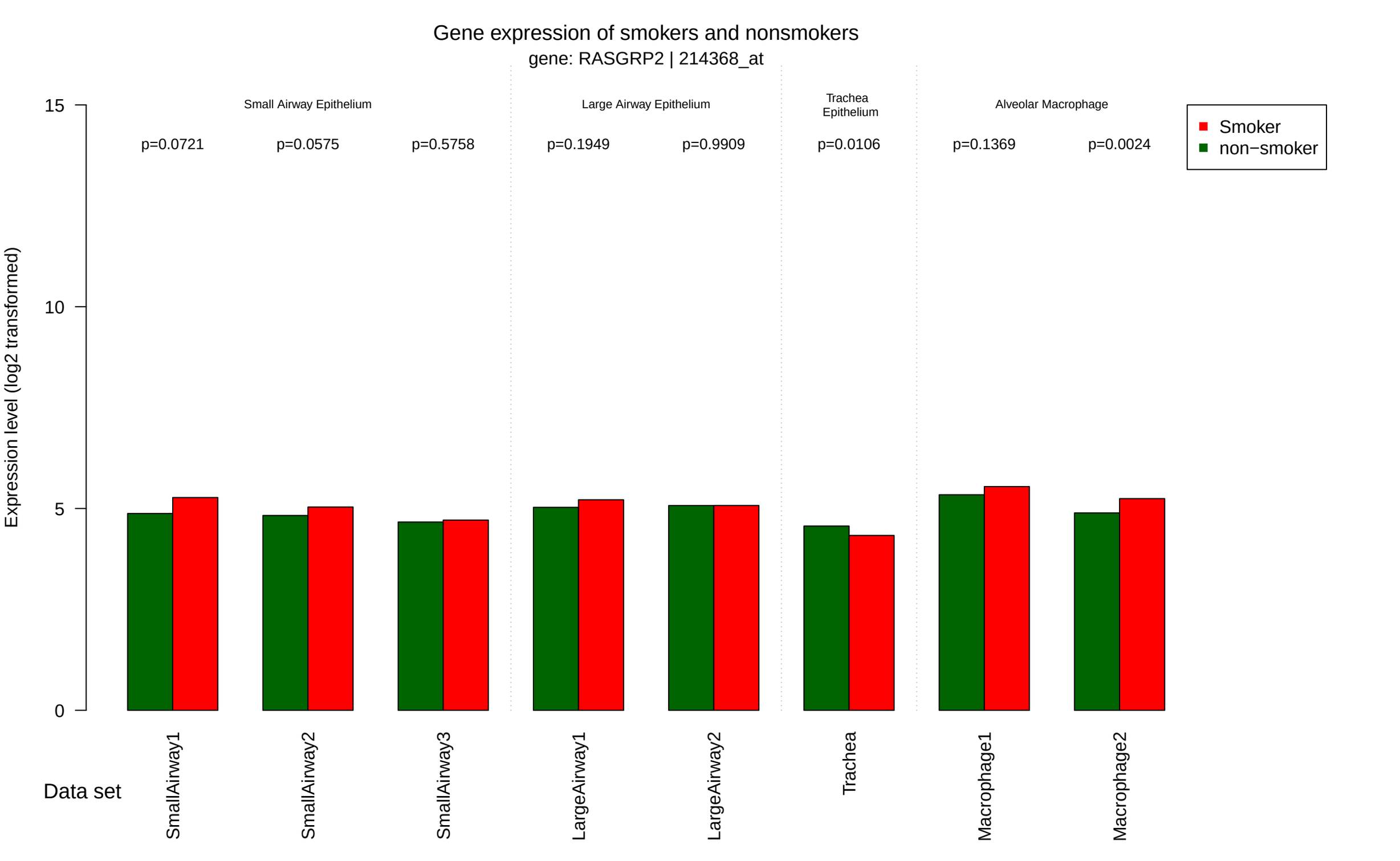
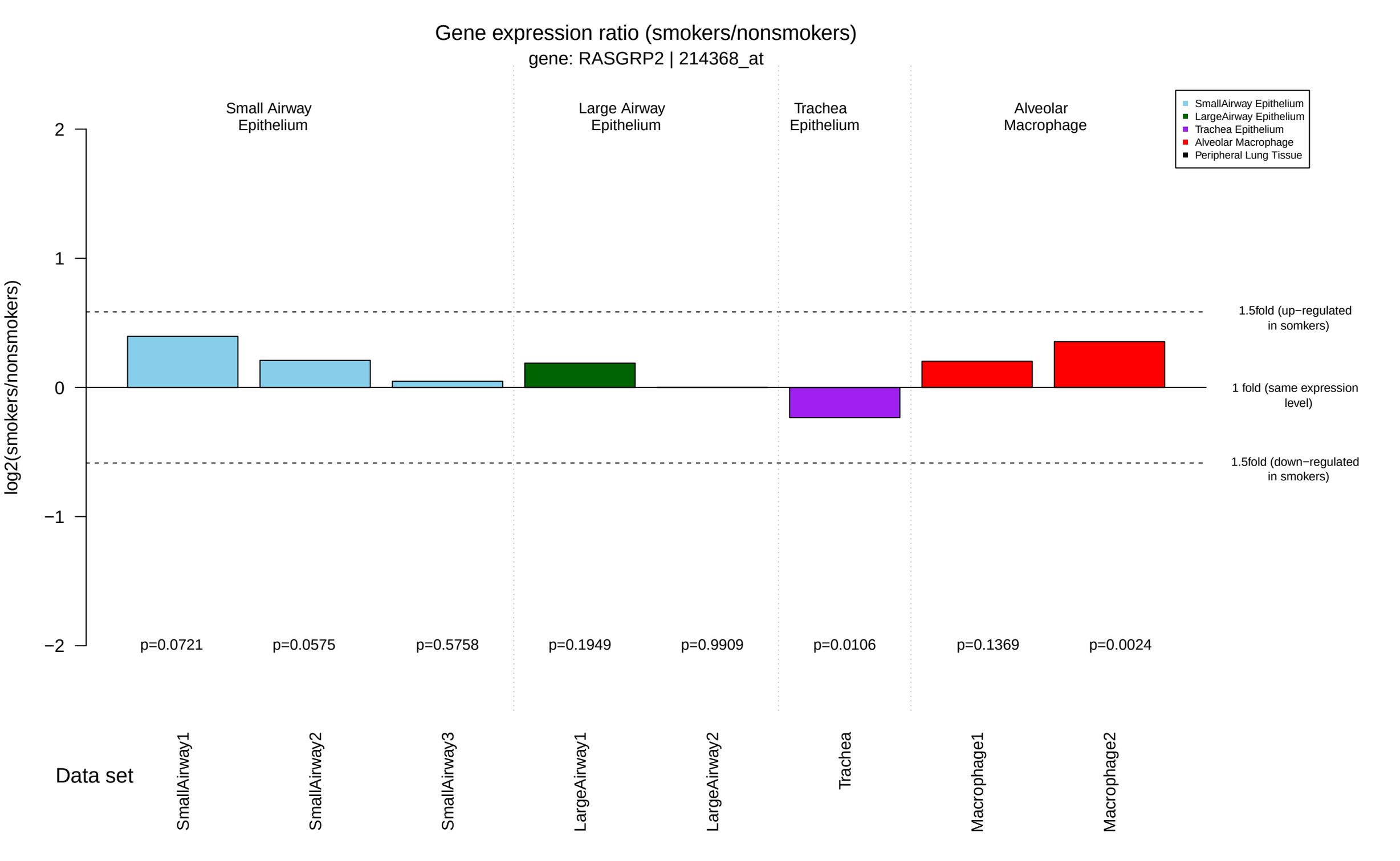
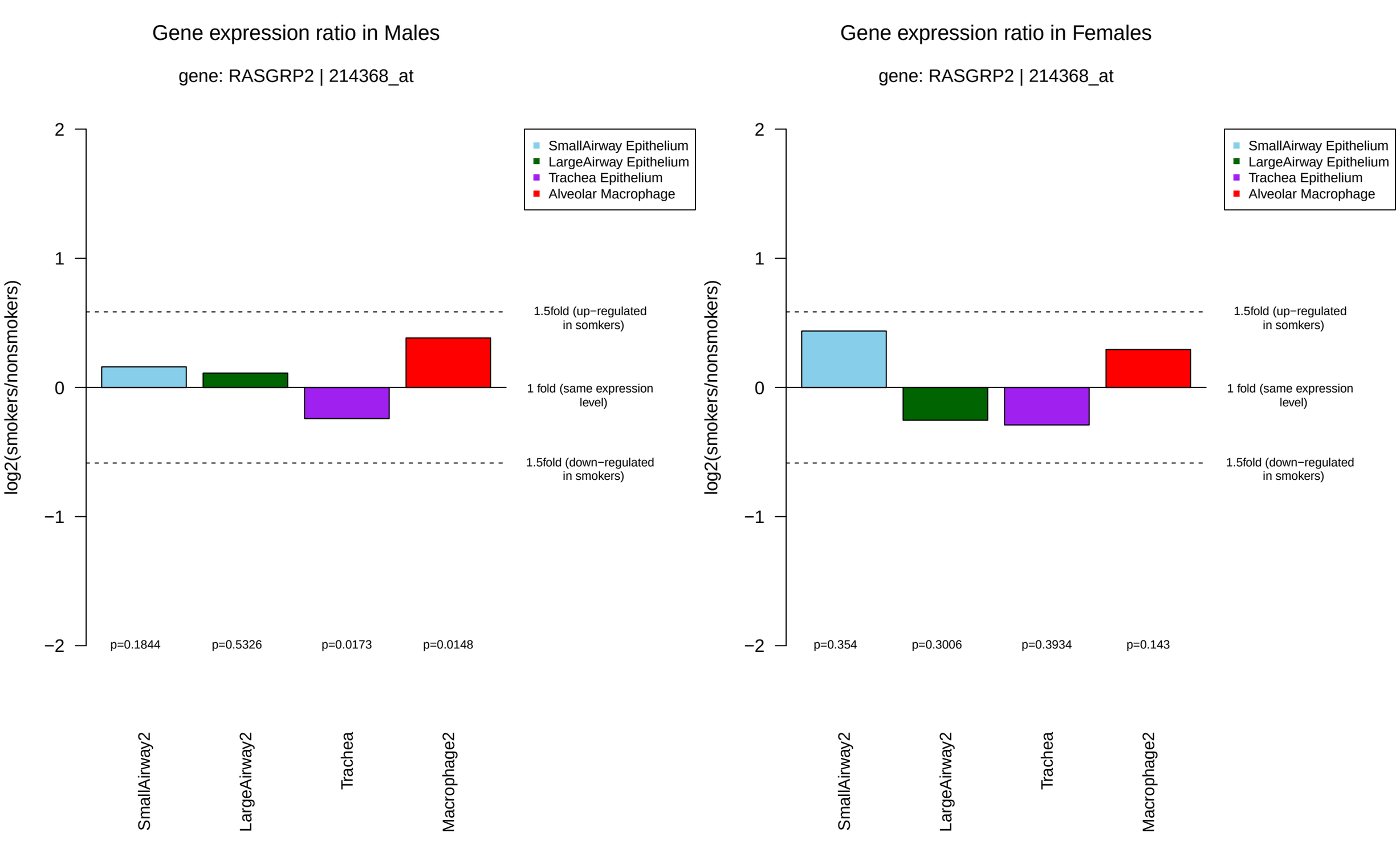
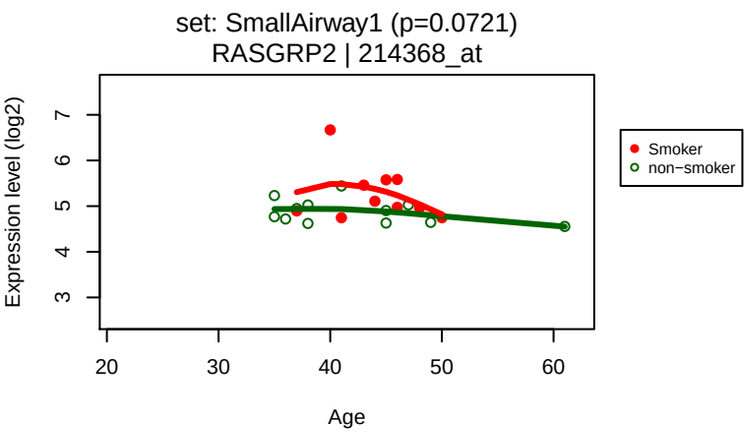
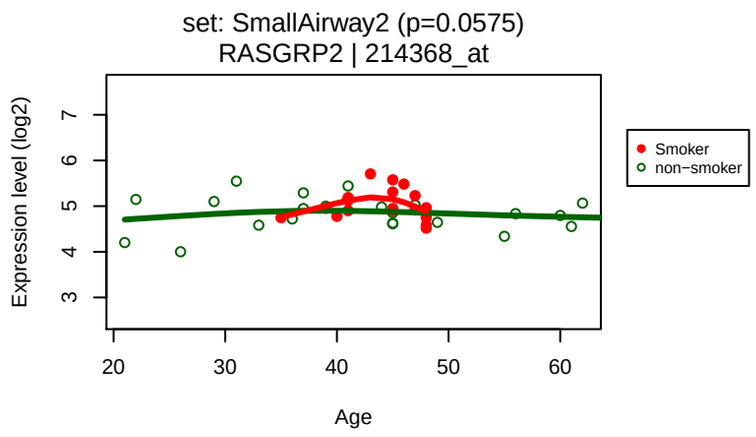
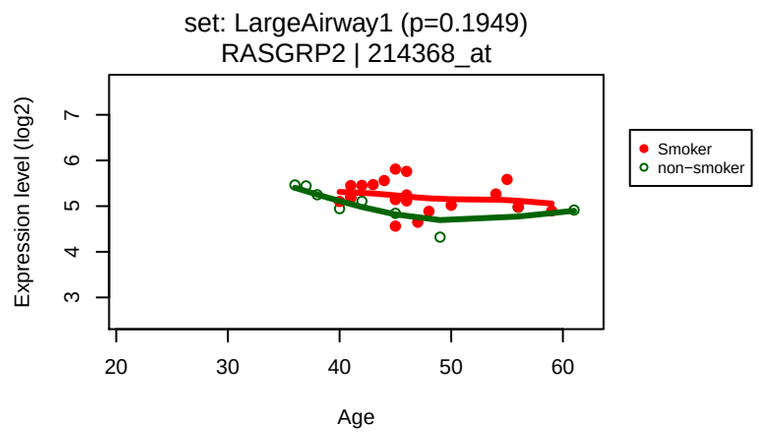

Gene expression of smokers and nonsmokers
gene: RASGRP2 | 214368_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0721
p=0.0575
p=0.5758
p=0.1949
p=0.9909
p=0.0106
p=0.1369
p=0.0024
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

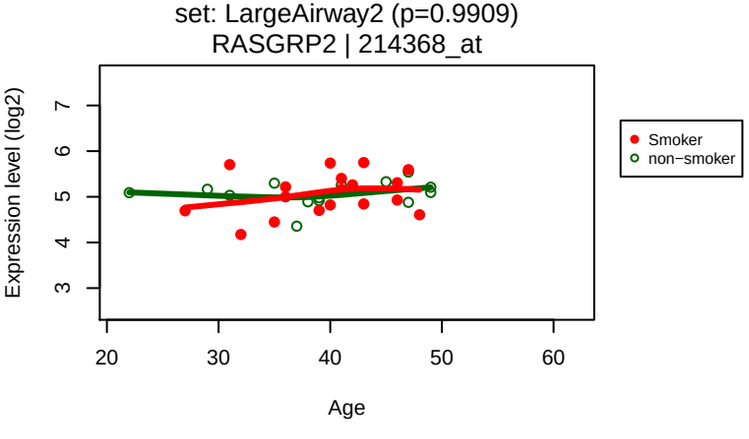

Gene expression ratio (smokers/nonsmokers)
gene: RASGRP2 | 214368_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0721
p=0.0575
p=0.5758
p=0.1949
p=0.9909
p=0.0106
p=0.1369
p=0.0024
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

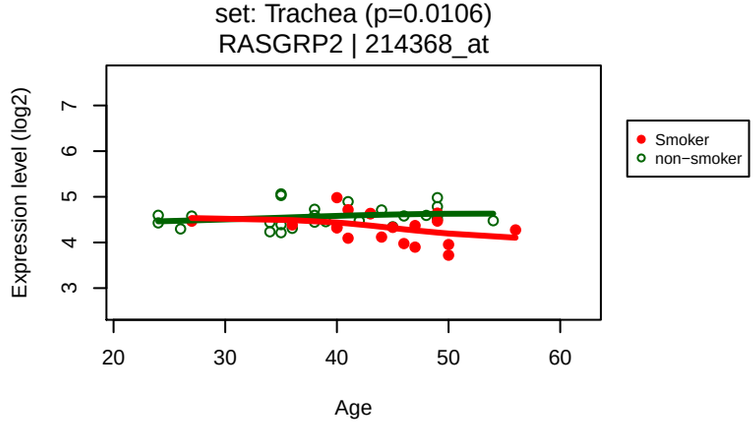

Gene expression ratio in Males
gene: RASGRP2 | 214368_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1844
p=0.5326
p=0.0173
p=0.0148
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RASGRP2 | 214368_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.354
p=0.3006
p=0.3934
p=0.143
SmallAirway2
LargeAirway2
Trachea
Macrophage2

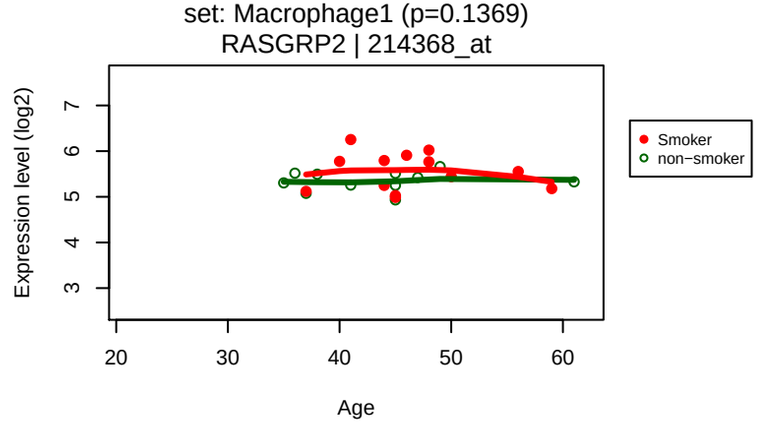

set: SmallAirway1 (p=0.0721)
RASGRP2 | 214368_at
Expression level (log2)
Age
Smoker
non−smoker

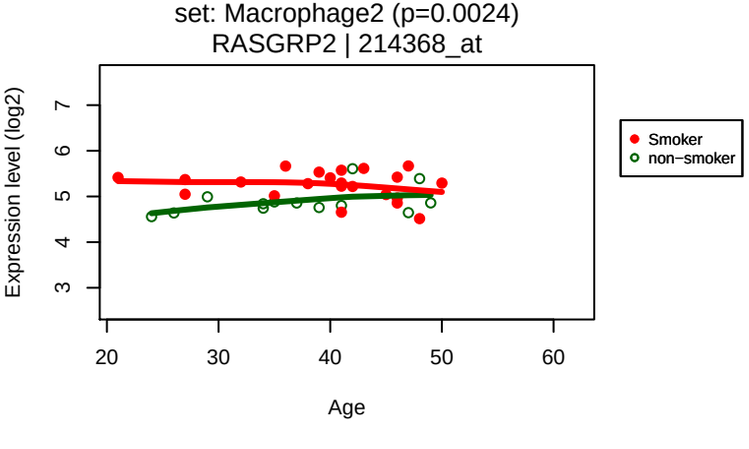

set: SmallAirway2 (p=0.0575)
RASGRP2 | 214368_at
Expression level (log2)
Age
Smoker
non−smoker

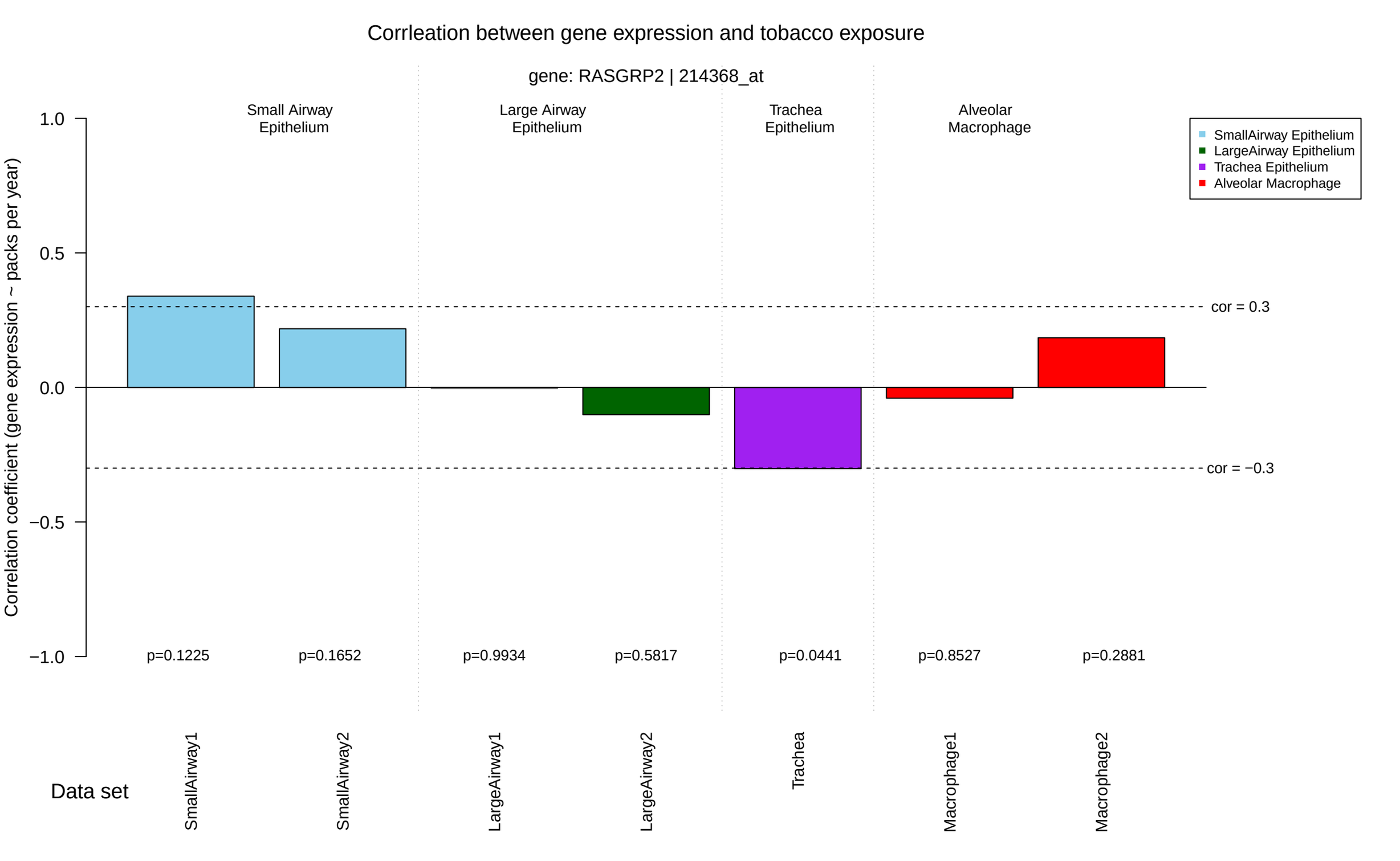

set: LargeAirway1 (p=0.1949)
RASGRP2 | 214368_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9909)
RASGRP2 | 214368_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0106)
RASGRP2 | 214368_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1369)
RASGRP2 | 214368_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0024)
RASGRP2 | 214368_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RASGRP2 | 214368_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1225
p=0.1652
p=0.9934
p=0.5817
p=0.0441
p=0.8527
p=0.2881
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2